Benjamin H. Roth

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Second-year Aerospace Engineering student and student pilot at Texas A&M University with a strong foundation in engineering. Seeking a full-time summer 2027 internship at any location in the aerospace and aeronautical industry to apply hands-on technical and collaborative problem-solving skills.

Education:

Texas A&M University College of Engineering: GPA: **3.6**
Bachelor of Science in Aerospace Engineering *Expected* **May 2029**
Involvement: NCWSA Aggie Waterski Team, American Institute of Aeronautics and Astronautics (AIAA) member

Skills:

Autodesk Inventor, Autodesk Fusion 360, XFLR5, Python, Aircraft Design, Aerodynamics, Flight Dynamics, Rapid Prototyping (3D Printing), Aircraft Maintenance, Flight Planning, Project Management, RC Pilot, Pilot (PPL w/45 Hrs)

Certifications:

Certiport Autodesk Inventor and Revit, REC Foundation Fundamentals of Engineering (w/ Electrical, Mechanical, and Programming Sub-Tests), Certiport Python, Florida Engineering Society: Engineering Core Certification

Experience/Projects:

Island Tyme Cargo LLC. at KLNA - Mechanical/Engineering Intern: **June 2026 - August 2026**

- Performed hands-on **airframe and powerplant maintenance** on Beechcraft Model 18 aircraft, logging practical hours toward my **FAA A&P certification**
- Conducted heavy maintenance and overhauls on supercharged Pratt & Whitney R-985 Wasp Junior radial engines, including cylinder replacements, complete engine removals/installations, spark plug inspections, and routine fluid services
- Aided in airframe repairs, including replacing flight control cables and rewiring systems, while supporting ground and flight-line logistics for cargo routes to the Bahamas

Cleva Technologies LLC. in Boca Raton, FL - Intern: **June 2024 - July 2024 & June 2025 - July 2025**

- Assisted in manufacturing and testing industrial fans while shadowing engineers and business managers to gain hands-on experience in real-world engineering, manufacturing, and design efficiency
- Developed hands-on **workshop** and **electrical** skills while contributing to the assembly of the Vector Fan chassis and supporting testing and quality evaluation

N422XS Inc. at KHSA - Seasonal Job: **Lifetime**

- Assisted engineers and A&P mechanics various times throughout my life in restoring a Cold War–era English Electric Lightning TMK5 supersonic fighter jet to flight

“Aggie Bullet” AIAA Design Build Fly - Stability and Control Subteam Member: **August 2025 - May 2026**

- Assisted in evaluating and optimizing aircraft configurations, including delta-wing layouts and different NACA airfoils, analyzing wing geometry and stability controls for Texas A&M's entry in the annual AIAA Design/Build/Fly (DBF) competition
- Assisted in validating aerodynamic performance and control authority across multiple flight missions through **XFLR5 analysis and experimental wind-tunnel testing**
- Developed custom **Python** scripts to process flight-test data, conduct performance trade studies, and model competition flight profiles to maximize overall mission score

Eagle Scout Service Project for Florida Atlantic University: **Summer 2025**

- Directed an end-to-end service build at FAU Pine Jog Environmental Center, managing site safety, material transport, and the fabrication of 12 weather-resistant wooden benches with a water-safe storage shelter
- Coordinated on-site structural building and removed hazardous debris using power and machining tools, which allowed me to gain further experience in **workshopping, carpentry, and design**
- Raised over $1200 through donations for the project and over 200 hours **managing and coordinating** the project

Personal Glider Project: **Summer 2026**

- Designed and built a remote-controlled, powered Glider in **XFLR 5** and conducted an **aircraft stability analysis**
- Conducted **sideslip analysis** and checked for key flight stability components at both 88 ft/s and 177 ft/s, such as lift production, stall margins, induced drag growth, etc through their respective graphs
- Flew the glider and compared the results to aircraft stability, made modifications such as removing the h-stab and switching it to conventional and adding **3d printed** winglets

SEA Mate Underwater Robotics - Team Engineer: **August 2024 - May 2025**

- Designed an underwater ROV to participate in the SEA Mate underwater robotics competition in Tampa, FL.
- Designed parts such as propellers, motor mounts, and other various parts for the rover in **Autodesk Fusion 360 CAD,** which were **3D printed** and **resin printed**
- Worked on electronics as well, including using **Python** to allow control of the ROV using an Xbox controller

Honors/Awards:

Haynes Scholarship Award, O.W. Stanley and Agnes Rabon Stanley Endowed Scholarship Fund, Reta Haynes Dean's Scholar Award, Nominated by Congresswoman Lois Frankel and Senator Rick Scott for USAF Academy, Eagle Scout